The pigeonhole principle

High school · olympiad thinking · Progressive exercises · 02

Exercise 1. Show that among 25 people, at least three were born in the same month.

Exercise 2. Choose six distinct numbers from $\{1, \dots, 10\}$. Prove that two sum to 11.

Exercise 3. Prove that among eight integers, two have a difference divisible by 7.

Exercise 4. Five points lie in a square of side 2, including its boundary. Show that two are at distance at most $\sqrt{2}$.